



# HOSPITAL SERIAN

**TITLE: DEPARTMENT OF PATHOLOGY AND BLOOD  
TRANSFUSION GENERAL OPERATIONAL POLICY**

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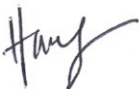
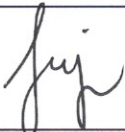
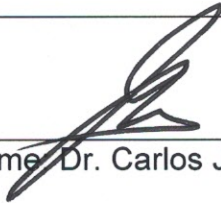
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## AMENDMENT PAGE

| Date of Amendment | Issues No / Amendment No. | Details of Amendment | Signature of Person who initiated the change | Signature of person who approved the change |
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## 1. INTRODUCTION

Pathology Department of Serian Hospital is a secondary level healthcare facility under the Malaysian Ministry of Health. As part of Serian Hospital, the department has been certified and meets the Malaysian Society of Quality in Health (MSQH) standards. Serian Hospital Pathology Department consists of Pathology and Blood Transfusion Services.

## 2. OBJECTIVES

- To provide efficient and good quality of laboratory services.
- To upgrade the services, equipment and procedure.
- To ensure and assure patient's confidentiality and comfort.
- To provide safe and continuous blood supply for those in need.

## 3. VISION

Pathology Department and Blood Transfusion is committed to cooperate in providing professional services for pathology analysis and quality blood supply.

## 4. MISSION

Pathology Department and Blood Transfusion aims to provide a comprehensive and quality laboratory services through innovative methods, efficient, cost-effective and latest technology.

## 5. ORGANIZATIONAL STRUCTURE

- The department is divided into 8 units (6 diagnostic and 2 non-diagnostic units). See attached organization chart (Appendix A).
- The State Pathologist and State Blood Transfusion Specialist is responsible for ensuring all policies and procedures are implemented in all laboratories and blood bank in MOH hospitals under his care, monitor the Quality performance and take appropriate action, coordinate training and propose required resources. They are also the technical advisors for the laboratory and blood bank services in the health clinics in their respective states. They are also expected to arrange regular meetings at state level.
- Serian Hospital laboratory headed by a Scientific Officer in the absence of a resident Medical Officer trained in laboratory services.

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- d) Supervisory visits to hospital without Pathologists / Blood Transfusion Specialist like Serian Hospital shall be conducted by the state Pathologist / Blood Transfusion Specialist or designee periodically.

## 6. STATISTIC

Data collection for statistical analysis shall follow the latest guidelines given by Ministry of Health Malaysia.

## 7. GENERAL POLICY

- a) The Department of Pathology provides diagnostic services in the following disciplines of pathology:
  - Chemical Pathology (Biochemistry)
  - Hematology
  - Medical Microbiology (Bacteriology, Serology, Virology, and Parasitology)
  - Blood Banking
- b) Apart from providing diagnostic services for Serian Hospital, the Pathology and Blood Transfusion Department also extends its diagnostic services to private practitioners as well as the Maternal & Child Healthcare (MCH) and Health Clinic within Serian vicinity.
- c) Regionalization or centralization of services for certain tests are based on tests that have low workload and are specialized, expensive and need highly skilled laboratory personnel.
- d) Out-sourcing of the tests shall be between government laboratories. If this is not possible the out-sourcing of services should be arranged with accredited or technically competent private laboratories. The laboratory responsible for out-sourcing according to a standardized or national Pathology outsourcing policy.
- e) Planning for new tests shall notify and get approval from the State Pathologist / Blood Transfusion Specialist and the Heads of the relevant Disciplines.
- f) Point of Care Testing (POCT) services shall be implemented in consultation with the Scientific Officer in charge of the laboratory. The laboratory shall take active role in monitoring quality performance involved in these services.
- g) The Blood Banking Unit organizes scheduled blood donation campaign to ensure continuous supply of blood for patient care.
- h) Maintain communication and provides consultative services to medical profession and other relevant staff in selection of laboratory investigation, interpretation and repeat test or sampling if required.

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- i) Apart from offering diagnostic services, the department also acts as a training centre for various levels and categories of staff including Medical Laboratory Technologies and students from institutes of higher learning.

## 8. SERVICE HOURS

- a) The Department of Pathology and Blood Transfusion provides 24 hours service for certain tests in Biochemistry, Haematology, Medical Microbiology and Blood Bank (APPENDIX D). A medical laboratory technology on call is available for assistance after office hours.
- b) The Out Patient Department (OPD) Specimen and Blood Taking Counter Unit operating time:  
Monday – Thursday: 07.30 a.m. – 4.30 p.m.  
Friday: 07.30 a.m. – 11.30 a.m.;  
2.15 p.m. – 4.30 p.m.  
Weekends and Public Holidays: 8 a.m. – 2 p.m.

## 9. OPERATIONAL PROCEDURES

### 9.1 Pathological Services

#### 9.1.1 Pre - analytical

- a) All tests shall request by using the test request form PER-PAT-301 (excluded special tests, e.g. Drug Screening tests, DNA Analysis and other special tests.)
- b) All tests requested shall provide the following information in the request form:
- Identification of the patient (full name, identity number, date of birth, sex)
  - relevant medical history of the patient
  - relevant medication of the patient (if applicable)
  - identification of the nature of the specimen
  - clear labelling of specimens requiring precautionary handling
  - Test of specimen requested
  - Date of specimen collected
  - Sign and chop from requesting specialist / medical officer / medical assistant
- c) Verbal requests for additional tests are discouraged, however they should be considered on a situational basis and depend on the stability of the analytes. These requests shall be documented and followed with written confirmation using the suitable request form if needed.
- d) General Workflow for Handling of Specimen in Pathology and Blood Transfusion Department, see attached workflow (Appendix B).

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- e) The samples/specimens should be collected by the ward or clinic staff, properly labeled and dispatched to the laboratory with appropriate containers as specified. However, for outpatients, their samples and specimens will be collected by the Medical Laboratory Technicians or Nurses on duty at the Out Patient Department (OPD) Specimen and Blood Taking Counter Unit.
- f) The specimen containers from wards / clinics / other departments must be labeled with at least two identifiers (i.e. name of patient and patient's I.C number) and test requested.
- g) The specimen containers should be placed in biohazard plastic bags with the respective request forms placed in the pocket of the biohazard plastic bags.
- h) Requestor shall clearly identify the type of tissue that removed in an operation, and the details of the patient shall have written clearly in the request form.
- i) The specimens should be sent to the laboratory together with a dispatch book or specimen lists and receive acknowledgement from the laboratory staff.
- j) URGENT samples must be brought to the laboratory by the ward or clinic staff immediately instead of using regular porter. The time of specimens received at the counter should be clocked in by the laboratory staff and the specimens will subsequently be attended by the laboratory staff for the URGENT tests to be carried out. For URGENT request after office hour, clinical requestor should call and inform on-call MLT.
- k) Urgent request forms must be signed either by a Medical Officer / Specialist / Medical Assistant depending on the category of tests requested.
- l) In case if there is any delay on testing should notify the in-charge medical officer or related clinical staffs immediately.
- m) Pathology and Blood Transfusion Department has the right to reject a request which do not fulfill the laboratory requirements as follows:
  - No specimen/requisition
  - Incorrect anticoagulant/preservative
  - Particular on specimen container and requisition form are not identical
  - Unknown source of specimen
  - Specimen unlabeled
  - Specimen clotted/grossly haemolyse
  - Slides broken/specimen leaking & spillage
  - Insufficient quantity of specimen
  - Illegible information/irrelevant or no clinical information
  - Doctor's (requesting Specialist / Medical Officer) chop or signature is not available

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- Date/time of specimen of specimen collection is not indicated.
- Sample is unsuitable for test requested.

n) Sample rejection shall be done through telephone or rejection form to inform the ward or clinic staff.

### 9.1.2 Analytical

- All new methods shall be validated prior to use and all data record are kept.
- The Laboratory must participate in External QA Programme. When QA programme is unavailable, the national/state Pathology heads shall coordinate inter laboratory comparison programs.
- Clinical interpretation of test result/report shall only be made by clinically qualified personnel (trained MO or Pathologist or Blood Transfusion Specialist).

### 9.1.3 Post Analytical

- Laboratory results can be collected by ward or clinic staff or can be obtained at the respective pigeon holes. All test results exceeding "critical values" that are established at national level will be informed to the requestor within 30 minutes after results validation and recorded in critical value record book.
- All URGENT tests are to be given immediate attention, and results to be dispatched within the LTAT established at National Level.
- The responsibility of any results informed or received through telephone or similar means will be delegated to the responsible staff. A hardcopy of the reports shall follow with minimal delay if possible.
- There is different retention duration for each type of specimens and patient results. See attached table (Appendix C)
- The statistics of workload by tests will be maintained following the guidelines set at national level. Annual statistics are to be compiled by the State Pathologists and sent to the National Pathology Advisor.

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## 9.2 Blood Bank Services

### 9.2.1 Blood Transfusion

- a) All blood transfusion shall request by using the form PER-SS-BT 105 / 88.PK5-PPAK5 by registered medical practitioner and consent of patient shall be obtained.
- b) Samples or materials examined shall be stored 3 days after test is done.
- c) Any blood and blood products that has not been stored correctly according to the standard guideline shall be discarded.
- d) All blood found to be positive of any of the diseases shall be discarded according to the standard procedures.
- e) The units of blood shall be issued/ used on first in first out basis according to the blood group.
- f) All units of blood requested from other Blood Bank shall be collected immediately (urgent case) or within 48 hours (non-urgent case) after the units of blood is ready to be collected.
- g) Prior administration of blood and blood component, all information shall be verified according to the checklist in national guideline.
- h) Transfusion therapy is under responsibility of registered medical practitioner. Only registered medical practitioner is authorized to prescribe and request for blood transfusion.
- i) Signature and name of the staff who performed the administration of blood product.
- j) All patients receiving transfusion shall be monitored before, during and after the transfusion process.
- k) All near misses and adverse events shall be managed and reported to Hospital Transfusion Committee and National Haemovigilance Coordinating Centre.
- l) Information of blood received or issued for transfusion must be recorded according to standard procedure.
- m) The Blood Transfusion Unit shall maintain the communication and provide consultative services to clinician and relevant staff on blood products and repeat of tests or sampling if required.

### 9.2.2 Blood Donation

- a) All blood donors shall have their hemoglobin, blood pressure and weight checked before their blood is taken.

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- b) The issuance of time slip shall be given only when requested by the donor.
- c) All blood donation shall be on voluntary, non-remunerated and no payment is allowed. Donor's consent shall be obtained before the donation take place.
- d) All blood collected shall be screened for Syphilis, HIV, Hepatitis B and Hepatitis C. All screening shall be sent within 72 hours after mobile blood donation and done in Blood Bank of SGH.
- e) Unscreened blood shall be stored separately from the screened blood.
- f) Only screened blood shall be used for transfusion.
- g) All blood donors are to be certified fit by the Medical Assistant or Medical Officer prior to blood donation.
- h) Medical Officer or Medical Assistant is to accompany all mobile blood donations to examine blood donors to ascertain they are fit to donate blood for that time.
- i) All blood donors whose blood is found to be positive for diseases shall be informed and referred to the medical officer, who is the committee member of Hospital Hemovigilance Committee (HTC) Serian Hospital. Hence the donors are deferred temporarily or permanently.
- j) Directed blood donation by family members is not recommended except in special circumstance (please consult Blood Transfusion Specialist).

## 10. FUNCTIONAL AREAS

### 10.1 Facilities & Equipment

#### 10.1.1 General

The laboratory is equipped with the minimum standard list of equipment based on the level of hospital and workload.

#### 10.1.2 Procurement

- a) Procurement of equipment can be made through direct purchase, reagent rental arrangement or donation following the policies and procedures of MOH.
- b) All proposed procurements shall be submitted to the Head of department through the State Pathologist and Director Serian Hospital.

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- c) All equipment procured shall meet the specifications for standard methodology used for the test to be performed as determined by the State Head of Discipline.
- d) Equipment for replacement shall fulfill the criteria for BER.
- e) Proper testing and commissioning of all newly acquired equipment must be conducted in the presence of the end-user, support service personnel and supplier, after which the equipment must be entered into the hospitals' assets list.

### 10.1.3 Maintenance

All equipment shall have planned preventive maintenance as a minimum following the manufacturer's recommendation.

### 10.1.4 Training

The vendor shall provide training and ensure competency of the laboratory staff before putting the equipment for routine use.

## 11. HUMAN RESOURCES

- a) Distribution of human resources depend on the service Departmental Policy of Pathology and Blood Transfusion Services in Ministry of Health Malaysia needs based on geographic location, subspecialty distribution and workload.
- b) The laboratory personnel shall be trained as a team for any new service/ program approved by MOH for the clinical discipline or patient care is concerned.
- c) Competency/performance assessment for all technical staff shall be conducted at least once in three yearly.
- d) Credentialing and privileging special procedures for all technical staff shall be in place.
- e) Post graduate and post basic training for all categories of staff shall be in accordance with service needs.

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- f) Only appropriately qualified staff shall be allowed to perform laboratory work that affects the quality of patient's results.
- g) All hospital MLTs are required to perform after hour's duty irrespective of their discipline.
- h) SO shall carry out after hour's duty when required.

## 11.1 Personnel

- a) Head of Department - Pathology laboratory and Blood Transfusion in Serian Hospital headed by a Scientific Officer under the supervision of a visiting State Pathologist and Blood Transfusion Specialist.
- b) Quality Manager – Scientific Officer or trained MLT to manage the Quality Management System (QMS) in the laboratory.
- c) Laboratory Manager – In Serian Hospital, the HOD of Pathology laboratory and Blood Transfusion acts as a Laboratory Manager.
- d) Scientific Officer – Only two SO posted in Serian Hospital, one Biochemist and one contract Microbiologist.
- e) Medical Laboratory Technician (MLT) – consist of U32 and U29.
- f) Attendant Hospital / *Pembantu Perawatan Kesihatan (PPK)* – function as personnel for manage consumables stock (as store keeper) in laboratory, sterilization of reusable container, transporting outsource specimens, dispatch of results / other documents and any other extended duties with training.

## 11.2 Training

### 11.2.1 General Policy

- a) All categories of staff (SO, MLT and PPK) in Pathology and Blood Transfusion service should be given opportunity to attend relevant courses during their service period in order to maintain efficiency and competency.
- b) All personnel shall be appropriately trained via pre-training exposure and scheduled educational program.
- c) The training of SO and MLTs shall be relevant and should meet the current demand of the service.

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- d) Credentialing and privileging program shall be implemented for special procedures (examples venipuncture procedure and local anesthetic).

### 11.2.2 Specific Policy

#### a) Policy for Scientific Officers (SO)

All newly employed Scientific Officers shall undergo training at state hospital before posting to peripheral hospitals. The training should encompass laboratory management which includes techniques, stringent quality control protocols, budget and equipment procurement.

#### b) Policy for Medical Laboratory Technologists (MLTs)

The training for newly employed MLT should pass laboratory management which includes techniques and stringent quality control protocols.

## 12. QUALITY ACTIVITIES

### 12.1 Quality Policy

It is our policy to provide reliable and timely result to our patients, we carried out laboratory testing and activities in accordance of the documented procedures and all the laboratory personnel is required to adhere to the policies and procedures in the Universal Standard Precaution and associated documents at all times.

### 12.2 Quality Improvement Activities

- a) The Quality Assurance Program is a daily activity in all laboratories throughout the State. All procedures of tests performed are imbued with quality assurance and quality control aspects towards the monitoring of results and ensuring that all results obtained are correct and reliable. Serian Hospital Laboratory also participated in the external Quality Assurance Program organized by the following organisations:

- Institute of Medical Research, Kuala Lumpur
- Pathology Department, General Hospital Kuala Lumpur
- Blood Transfusion Center, Kuala Lumpur
- Vector Control Department, Sarawak State Health Department

- b) Participation in external Quality Assurance activities involved the following:

- External Quality Assurance Clinical Chemistry Monthly Programme (EQAS)
- External Quality Assurance (EQA) for Sputum (Medical Microbiology)
- Quality Control for BFMP (Parasitology)

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- National Quality Assurance Program for bacteriology (Medical Microbiology)
- National External Quality Assurance (NEQABB) for Blood Bank
- External Quality Assurance Hematology Programme (EQAS)
- National Indicator Assurance (NIA) programmes
- Hospital Performance Indicator Accountability (HPIA)

- c) Apart from the abovementioned quality improvement activities, we also conduct customer satisfaction surveys twice a year, reviewing incident reports (if any), management review meetings whereby appropriate measures to curb problems as well as to ensure if any corrective actions have been taken.
- d) All patient and healthcare information shall be confidential in all performance improvement activities.

## 13. TRANSPORTATION

### 13.1 Organizational Policy

The transportation of clinical samples shall be the responsibility or monitor by Pathology and Blood Transfusion Services.

### 13.2 Scope of Service

- a) The scope of services shall encompass from collection of specimens until results are received by sender. It includes intra Laboratory (unit to unit in the same laboratory), Intra Hospital (sampling to the Lab.) and Inter Hospital (Outsourcing sample to other hospital).
- b) The type of service chosen, ranging from human, mechanical, to dedicated Pathology transport or outsourcing courier service, depend on "best fit" and available resources.

## 14. SAFETY MEASURES & PRECAUTIONS

Strict safety measures are practiced in accordance to the Standard Universal Precautions and associated Laboratory Safety Manual especially during handling of the specimens and chemicals.

Chemical disposal done periodically or when necessary. All the chemical disposal will be managed by professional vendor through Hospital Service Company. A copy of CSDS / MSDS for each chemical shall be provided to the vendor upon collection.

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## 15. OUTSOURCING OF TEST

- The Pathology and Blood Transfusion Department will be outsourcing certain tests which are not offered by this laboratory.
- Transportation of specimen to the outsourced laboratory and handling of the results/reports will be managed by the Department of Pathology and Blood Transfusion.
- Pathology and Blood Transfusion Department has the right to reject a request if the requirements are not fulfilled.

## 16. HOSPITAL WIDE POLICIES

Relevant aspects of the whole Serian Hospital policies shall be observed and adhered.

## 17. REFERENCE

- Serian Hospital Departmental Operational Policies, Pathology Department (2018)
- Pathology Services Handbook 2009, Department of Pathology Hospital Kuala Lumpur
- Malaysian J Pathology 2005; 27(1) : 51 - 56
- Departmental Policy of Pathology Services, September 2010
- Jadual Pelupusan Rekod Perubatan 2016, Kementerian Kesihatan Malaysia*
- Transfusion Practice Guidelines, 4<sup>th</sup> Edition, 2016

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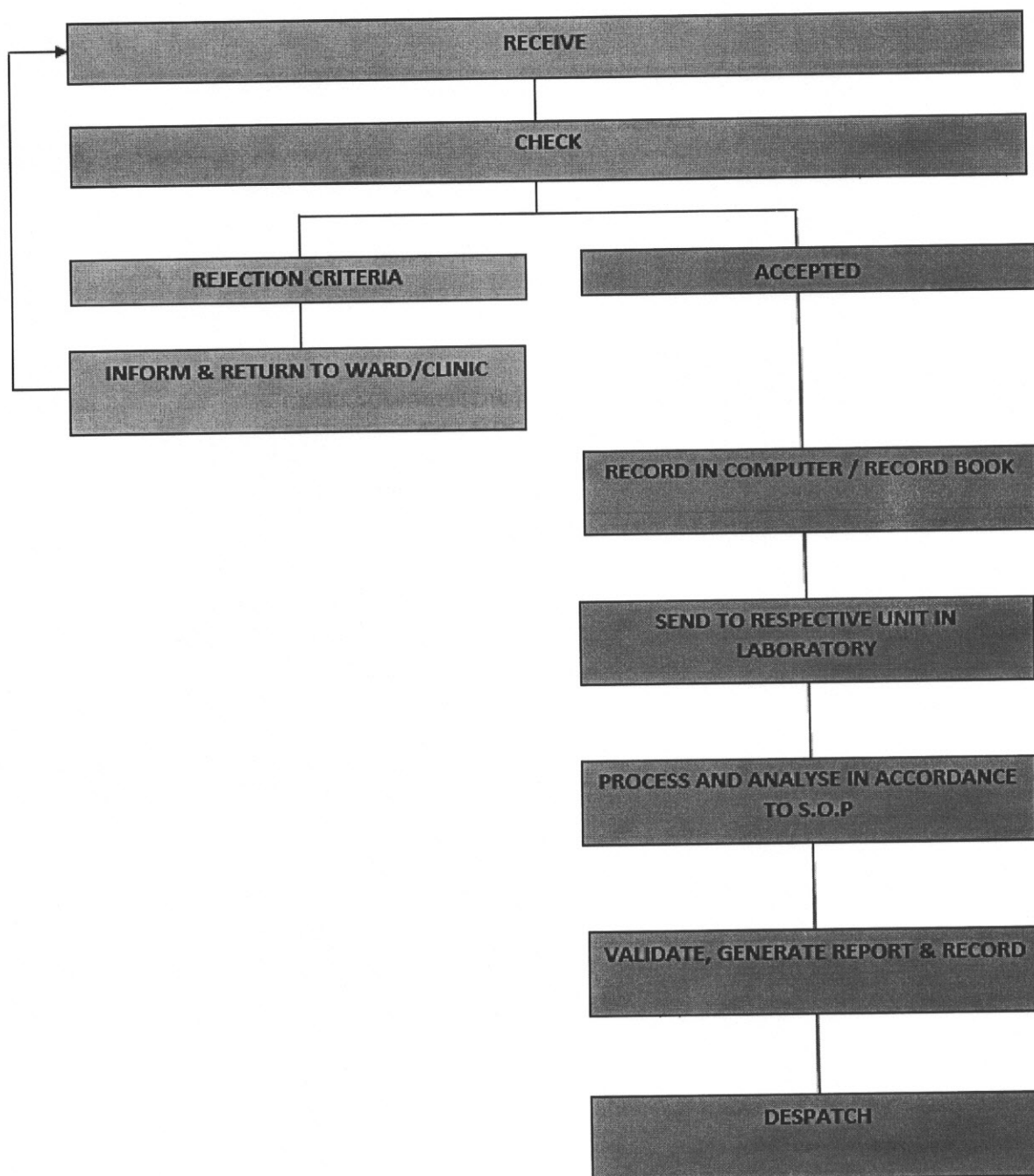
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## APPENDIX B

### GENERAL WORKFLOW FOR HANDLING OF SPECIMEN IN PATHOLOGY AND BLOOD TRANSFUSION DEPARTMENT



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## APPENDIX C

### RETENTION OF PATHOLOGY AND BLOOD TRANSFUSION RECORD AND SPECIMEN

|     | Record/Specimen                                           | Retention duration           |
|-----|-----------------------------------------------------------|------------------------------|
| 1.0 | General                                                   |                              |
|     | 1.1 Management                                            |                              |
|     | 1.1.1 Incident report                                     | Indefinite                   |
|     | 1.1.2 Training record                                     | 5 years                      |
|     | 1.1.3 Laboratory statistic                                | 5 years                      |
|     | 1.1.4 Duty roster                                         | 1 year                       |
|     | 1.2 All QC data and QA record                             | 3 years                      |
|     | 1.3 External QA end-cycle- summary                        | 5 years                      |
|     | 1.4 Remedial action log                                   | 5 years                      |
| 2.0 | Equipment                                                 |                              |
|     | 2.1 Maintenance, service, repair and calibration record   | Life of equipment            |
|     | 2.2 Daily, weekly, monthly record                         | 6 months                     |
|     | 2.3 Temperature record                                    | 6 months                     |
| 3.0 | Biochemistry                                              |                              |
|     | 3.1 Results                                               | 7 years                      |
|     | 3.2 Report Duplicate                                      |                              |
|     | 3.2.1 Drug of abuse testing (confirmatory)                | 7 years                      |
|     | 3.3 Samples                                               |                              |
|     | 3.2.1 Blood samples, body fluid                           | 2 days after issue of result |
|     | 3.2.2 Urine                                               | Discard after issue result   |
| 4.0 | Hematology                                                |                              |
|     | 4.1 Results                                               |                              |
|     | 4.1.1 Full Blood Count                                    | 7 years                      |
|     | 4.1.2 Routine coagulation test                            | 7 years                      |
|     | 4.1.3 G6PD Screening                                      | 7 years                      |
|     | 4.1.4 Full Blood Picture                                  | 7 years                      |
|     | 4.1.5 Hb Analysis                                         | 20 years                     |
|     | 4.1.6 Special coagulation test (Factor assays, platelets) | 20 years                     |

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|                                                                                  | 4.2 Samples                                             |  |                                                                                |                                                                                     |     |              |
|                                                                                  | 4.2.1. Blood samples                                    |  |                                                                                | 2 days after test is done                                                           |     |              |
|                                                                                  | 4.2.2 Coagulation blood samples                         |  |                                                                                | 2 hours after issue result                                                          |     |              |
| 5.0                                                                              | Microbiology                                            |  |                                                                                |                                                                                     |     |              |
|                                                                                  | 5.1 Specimen for culture and sensitivity (petri dish)   |  |                                                                                | 2 days after issue result                                                           |     |              |
|                                                                                  | 5.1.2 Urine, Sputum, gastric lavage                     |  |                                                                                | Discard after issue result                                                          |     |              |
|                                                                                  | 5.1.3 Blood                                             |  |                                                                                | Negative - Discard after issue result<br>Positive – Discard after issue result      |     |              |
|                                                                                  | 5.1.4 Body Fluid                                        |  |                                                                                | 2 days after issue result                                                           |     |              |
|                                                                                  | 5.2 Serum/plasma for serology                           |  |                                                                                | Negative - Discharge after issue result<br>Positive – 5 days after issue of results |     |              |
|                                                                                  | 5.3 Results                                             |  |                                                                                | 7 years                                                                             |     |              |
| 6.0                                                                              | Blood Transfusion                                       |  |                                                                                |                                                                                     |     |              |
|                                                                                  | 6.1 Request form                                        |  |                                                                                | 20 years                                                                            |     |              |
|                                                                                  | 6.2 Laboratory records of blood donations & transfusion |  |                                                                                | 20 years                                                                            |     |              |
|                                                                                  | 6.3 Samples of materials examined                       |  |                                                                                | 3 days after test is done                                                           |     |              |
| 7.0                                                                              | Cytology                                                |  |                                                                                |                                                                                     |     |              |
|                                                                                  | 7.1 Results                                             |  |                                                                                | 20 years                                                                            |     |              |
| 8.0                                                                              | Histology                                               |  |                                                                                |                                                                                     |     |              |
|                                                                                  | 8.1 Results                                             |  |                                                                                | 20 years                                                                            |     |              |

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## APPENDIX D

### LIST OF TEST OFFER AFTER OFFICE HOURS (ON-CALL)

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>BIOCHEMISTRY</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <ol style="list-style-type: none"> <li>1. BUSE</li> <li>2. SERUM/PLASMA CREATININE</li> <li>3. FASTING/RANDOM BLOOD SUGAR</li> <li>4. SERUM BILIRUBIN (PAEDIATRICS)</li> <li>5. LIVER FUNCTION TEST</li> <li>6. SERUM/PLASMA CREATINE KINASE (CK)</li> <li>7. SERUM/PLASMA LDH</li> <li>8. SERUM/PLASMA AMYLASE</li> <li>9. SERUM/PLASMA CALCIUM</li> <li>10. SERUM/PLASMA PHOSPHATE</li> <li>11. SERUM/PLASMA MAGNESIUM</li> <li>12. CEREBROSPINAL FLUID (CSF) BIOCHEMISTRY</li> <li>13. PARAQUAT SCREENING TEST</li> <li>14. URINE PREGNANCY TEST</li> <li>15. URINE DRUG SCREENING TEST</li> </ol> |
| <b>BLOOD TRANSFUSION</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <ol style="list-style-type: none"> <li>1. CROSS-MATCHING</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>HEMATOLOGY</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <ol style="list-style-type: none"> <li>1. FULL BLOOD COUNT (FBC)</li> <li>2. PROTHROMBIN TIME (PT)</li> <li>3. PARTIAL THROMBOPLASTIN TIME (PTT)</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>MICROBIOLOGY &amp; PARASITOLOGY TESTS</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <ol style="list-style-type: none"> <li>1. CSF GRAM STAIN, CULTURE, INDIAN INK</li> <li>2. BLOOD FILM FOR MALARIA PARASITES</li> <li>3. NS1 COMBO</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                           |

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